UNIT 0 · THE TOOLBOX · SKILLS CHECK MINI-PROJECT · OPTION 2

The Cafecito Receipt

Computer Science Math 2026–27 · in class Thursday, September 10 · **45 minutes, start to finish**

NAME _____ DATE _____

The job. You run the register at a cafecito window on Calle Ocho. At close, you export the day's sales — and the export is a mess. Your manager wants one number the register never showed: **total revenue for the day, and which item carried it.** You get the data, you **clean it**, you **convert the money with a formula**, and you **pick the chart that actually answers the question.** This is a skills check, not a model: it checks the tools you already have — and nothing you haven't learned yet.

What you receive

ONE CSV · ABOUT A DOZEN ROWS

0 The sales export

MESSY ON PURPOSE

Four columns: `Item`, `Price`, `Qty`, `Tip %`. Most rows are fine. Several are not — that is the point. Expect to find, in no particular order:

- a few prices typed as text, complete with a dollar sign: `"$4.50"`
- trailing spaces in some item names
- **one row priced in euros** (it says €) and one row where `Qty` is blank
- one **duplicated row**
- one item written in a different case than the rest

You do not need VLOOKUP for this. Nothing here is a lookup — that tool belongs to the food tracker next unit. Today is about honest formulas, clean columns, and the right chart.

The build — step by step

BUDGET YOUR 45 MINUTES

STEP	TASK	TIME
1	Set up the sheet and the KDEA header	~5 min
2	Clean the data	~12 min
3	Convert and compute	~13 min
4	Choose and build the chart	~10 min
5	Label check and conclusion	~5 min

1 Set up the sheet

STRUCTURE · KDEA

One tab, top to bottom: a title, a **labelled inputs block**, the data table, a results block, the chart. Every column header carries a **unit** (`$`, `%`, `count`). At the top, run the protocol:

K	THE GIVENS: THE DATA, AND THE EXCHANGE RATE YOU WERE HANDED
D	TOTAL REVENUE IN DOLLARS · THE ITEM THAT CARRIES IT
E	THE CONVERSION AND THE FORMULAS (VALUE, TRIM, ×, SUM…)
A	CELL REFERENCES FIRST, LABELS ON EVERY COLUMN

The exchange rate gets its own labelled input cell (for example, `1 € = 1.08 $`). You will reference that cell — you will not type `1.08` into a formula.

2 Clean the data

THE PART THAT EARNS THE GRADE

a. **Text to number.** Prices with a `$` are text, not numbers. Retype them, or strip and convert with `=VALUE( ... )`. Watch the left-alignment — a number that sits left is a string.

b. **Trim the names.** Remove trailing spaces (`=TRIM( ... )`), and make the odd-cased item consistent.

c. **Convert the euro row,** using the labelled input cell — not a typed constant.

d. **Handle the blank Qty.** Decide (set it to `1`? flag it?), and **write down what you decided**.

e. **Remove the duplicate row** — and note that you did. A duplicate silently inflates the total.

Leave a short **"What I cleaned and why"** note on the sheet. Cleaning is part of the answer.

3 Convert and compute

FORMULAS, NOT NUMBERS

a. `Revenue = Price × Qty` — write it once, fill it down.

b. `Tip $ = Revenue × Tip %`, then `Total = Revenue + Tip $`.

c. Results block: `=SUM` total revenue · `=COUNT` number of sales · `=AVERAGE` price · `=MAX` biggest sale.

d. Check yourself: does the count match the number of items *after* cleaning (duplicate gone, blank handled)? If not, something is still wrong.

4 Choose the chart — and defend it

THE DECISION IS GRADED

You have two reasonable choices, and the question decides which:

- **Bar chart** — each item's revenue side by side. Best when the question is *"which item carries the day?"*
- **Pie chart** — each item's share of the total. Only defensible if there are *few* slices; a dozen slices is soup.

Build your pick with a **title, both axis labels, and units on the value axis**, then write one sentence: *why this chart and not the other.*

5 Conclusion

ONE OR TWO SENTENCES

State the total revenue, name the item that carried it, and say one thing that would change your number (a different exchange rate? the duplicate handled the other way?). A conclusion that only describes the chart is not a conclusion.

Where students trip up

READ THIS BEFORE YOU START

- **Adding text to a number.** A `"$4.50"` price contributes **0** to a `SUM` — silently. The total looks plausible and is wrong. Fix the type first.
- **Hard-coding the exchange rate.** Typing `1.08` inside a formula breaks formula honesty, and the total won't move if the rate changes. Reference the input cell.
- **Forgetting the euro row.** If one row is in euros and you sum them all, you have added euros to dollars. Convert first, then sum.
- **The duplicate.** Left in, it double-counts one item and quietly inflates both the total and the pie slice.
- **A pie with too many slices.** Past about five or six, nobody can compare arcs. The bar chart wins — say so.
- **Unlabelled chart.** No title or no axis units scores below proficient even when the math is perfect.
- **Money math and rounding.** Format prices to two decimals for display, but let the formula keep full precision — don't retype a rounded number.

What you hand in

ONE WORKBOOK

- One tab: title · KDEA header · labelled inputs · clean data · results block · chart · conclusion.
- A formula in every computed cell — **no typed answers**.
- Every column header carrying its unit; every chart axis labelled.
- Your “What I cleaned and why” note.

Scored on the Toolbox rubric: Model Correctness, Formula Honesty, Data Structure, Communication. The tracker — and VLOOKUP — come next.